

Palm Target Localization

1. Experiment Objective

Use MediaPipe Hands to detect the position of the palm in the camera image and output the center coordinates of the palm's bounding box.

2. Experiment Principle

1. Key Terms

Keyword	Description
Palm bounding box	The minimum bounding rectangle computed from the 21 hand landmarks that encloses the whole palm
Center coordinates	The geometric center of the bounding box (indexx, indexY), used as the palm position output
Normalized coordinates	Coordinates normalized to the 0~1 range, so they can be passed downstream as control values

2. Technical Architecture

```
Terminal 1: start agent + camera + joint models
Terminal 2: start palm localization
```

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☒ Supported
No-PTZ version	☒ Supported
Required peripherals	ESP32 camera

2. Start the agent + camera + joint models (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```

yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-10-47-55-372666-yahboom-149737
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [149778]
[INFO] [camera_driver-2]: process started with pid [149780]
[INFO] [robot_state_publisher-3]: process started with pid [149782]
[INFO] [joint_state_publisher-4]: process started with pid [149784]
[INFO] [ekf_node-5]: process started with pid [149803]
[camera_driver-2] [INFO] [1785206876.886150022] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[joint_state_publisher-4] [INFO] [1785206877.007751174] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[micro_ros_agent-1] [1785206877.173080] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785206877.754391885] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785206877.754890084] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785206877.754912957] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754918130] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785206877.754924428] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785206877.754929199] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785206877.754935071] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785206877.754939792] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785206877.754944035] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754949951] [robot_state_publisher]: got segment rfwheel_Link
[camera_driver-2] [WARN] [1785206879.871708745] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785206880.046238399] [esp32_ip_camera_publisher]: IP camera stream opened successfully

```

3. Start palm localization (Terminal 2)

```
ros2 launch microros_v2_mediapipe_pkg find_hand.launch.py
```

4. Operating Instructions

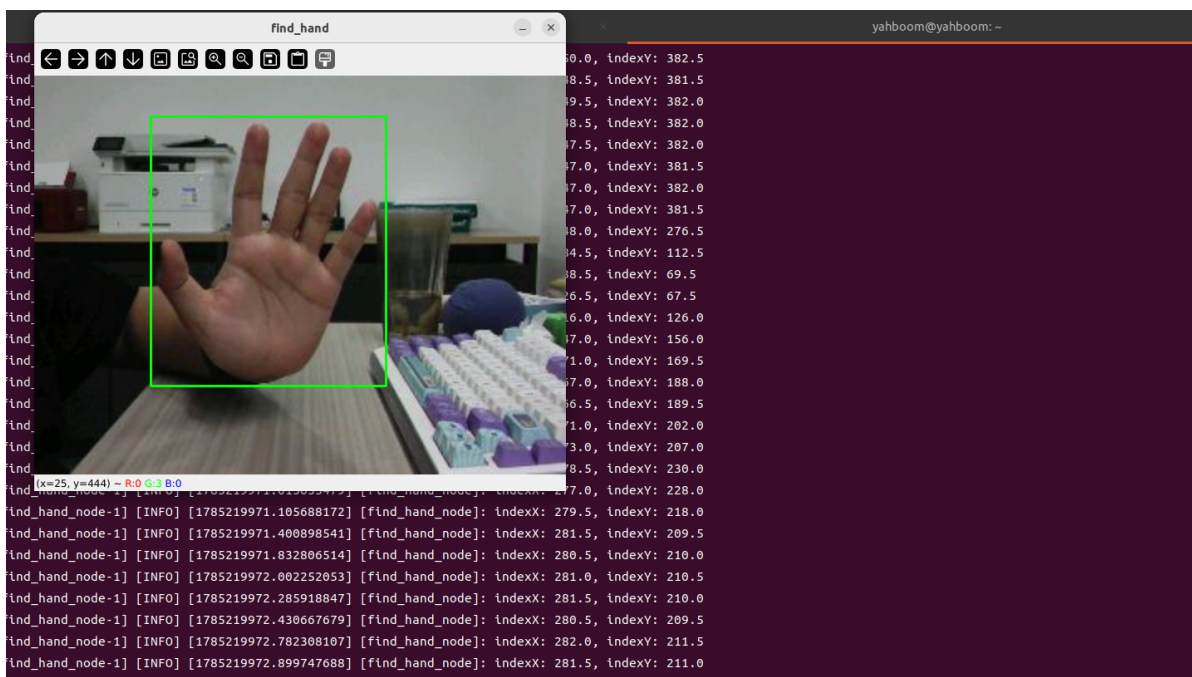
Place your palm in the camera's field of view. The screen shows the palm bounding box and a red center dot (indexx, indexY). Press **q** to exit.

5. How to Stop

Press **q** to exit.

4. Observed Results

- The palm is enclosed by a blue rectangular bounding box
- A red dot at the center marks the (indexx, indexY) coordinates
- As the palm moves across the frame, the coordinate values update in real time



5. Code Walkthrough

Source code paths:

- Launch:
`~/microros_ws/src/microros_v2_mediapipe_pkg/launch/find_hand.launch.py`
- Node:
`~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/FindHand.py`

1. Core Node Analysis

```
# Source file:
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/FindHand.py

class FindHand(Node):
    def __init__(self):
        super().__init__("find_hand_node")
        self.hands = mp.solutions.hands.Hands(
            static_image_mode=False, max_num_hands=1,
            min_detection_confidence=0.5, min_tracking_confidence=0.5)

    def image_callback(self, msg):
        frame = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
        results = self.hands.process(cv.cvtColor(frame, cv.COLOR_BGR2RGB))
        if results.multi_hand_landmarks:
            for hand_landmarks in results.multi_hand_landmarks:
                h, w, _ = frame.shape
                # Compute the palm bounding box from the 21 landmarks
                xs = [lm.x for lm in hand_landmarks.landmark]
                ys = [lm.y for lm in hand_landmarks.landmark]
                x_min, x_max = int(min(xs)*w), int(max(xs)*w)
                y_min, y_max = int(min(ys)*h), int(max(ys)*h)
                # Center coordinates (normalized)
                indexX = (x_min + x_max) / 2 / w
                indexY = (y_min + y_max) / 2 / h
                cv.rectangle(frame, (x_min, y_min), (x_max, y_max), (255,0,0),
2)
                cv.circle(frame, (int(indexX*w), int(indexY*h)), 8, (0,0,255),
-1)
                cv.putText(frame, f'({indexX:.2f},{indexY:.2f})', ...)
            cv.imshow(self.window_name, frame)
```

Key logic:

- The bounding box = the min/max of the 21 landmarks' x/y coordinates, not relying on MediaPipe's built-in rectangle (more flexible)
- Center coordinates normalized to 0~1: $(x_{min}+x_{max})/2/w$, so downstream control nodes can directly use the ratio value
- A red solid dot (radius 8px) + `(indexX, indexY)` text marks the palm position

2. Key Parameters

Parameter definition files:

- Launch:
`~/microros_ws/src/microros_v2_mediapipe_pkg/launch/find_hand.launch.py`

- Node:

```
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/Find  
Hand.py
```

Parameter	Type	Default	Description
<code>min_detection_confidence</code>	float	0.5	Detection confidence threshold
<code>min_tracking_confidence</code>	float	0.5	Tracking confidence threshold

3. Node Description

Node	Function
<code>find_hand_node</code>	MediaPipe Hands 21-point detection + palm bounding box + normalized center coordinate output

6. Frequently Asked Questions

Issue	Cause	Solution
Coordinates jump around	Palm detection is unstable between frames	Move the hand more slowly and keep the lighting uniform